RESEARCH PROCESS AND WAY OF THINKING

ROADMAP

Joy of Statistics

Research Process

Steps in a Research Study

Asking Good Questions

Operationalize

THE JOY OF STATISTICS

WHAT'S THE JOY IN STATISTICS, ANYWAY?

STATISTICS HELP US UNDERSTAND AND INTERPRET THE WORLD AROUND US

THEY PROVIDE INSIGHTS INTO COMPLEX PHENOMENA

STATISTICS EMPOWER US TO MAKE INFORMED DECISIONS

THE POWER OF VISUAL STATISTICS

- **Crimespotting (Stamen Design)**
  - Visualizes relationship between topography and crime
  - Empowers citizens with information for accountability
  - Showed the “promise of visualizations of public data”
- **Polar Area Graph (Florence Nightingale)**
  - Illustrated deaths from preventable diseases in hospitals
  - Sparked a revolution in hospital hygiene
- **Billion Pound-O-Gram (David McCandless)**
  - Helps comprehend large numbers in news and government reports
  - Puts financial figures into perspective
- **"We Feel Fine" Project (Jonathan Harris & Sep Kamvar)**
  - Explores human emotion on a global scale
  - Uncovers trends in collective feelings

- **Crimespotting (Stamen Design)**
  - Visualizes relationship between topography and crime
  - Empowers citizens with information for accountability
  - Showed the "promise of visualizations of public data"

e)

seases in hospitals

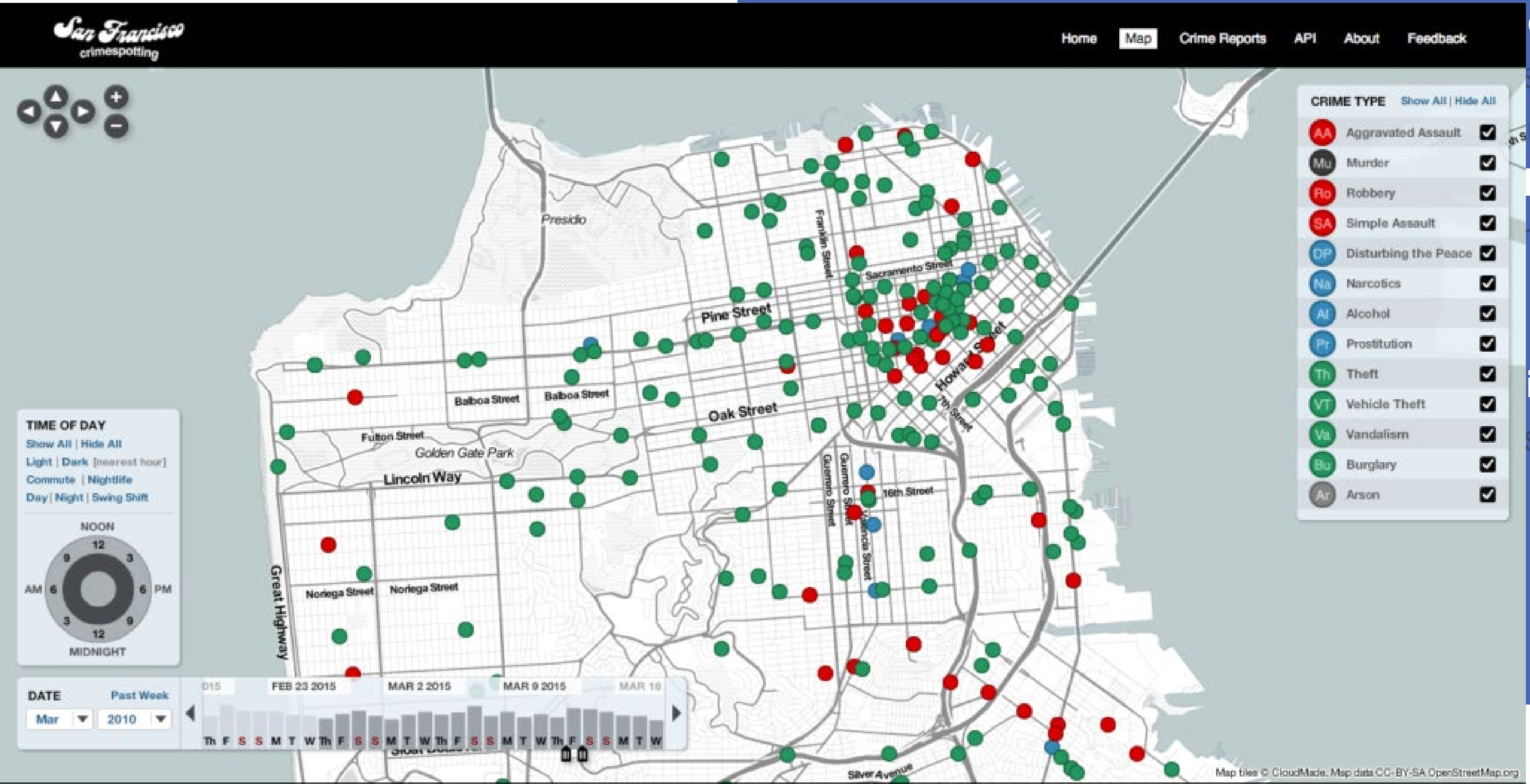
ne

less)

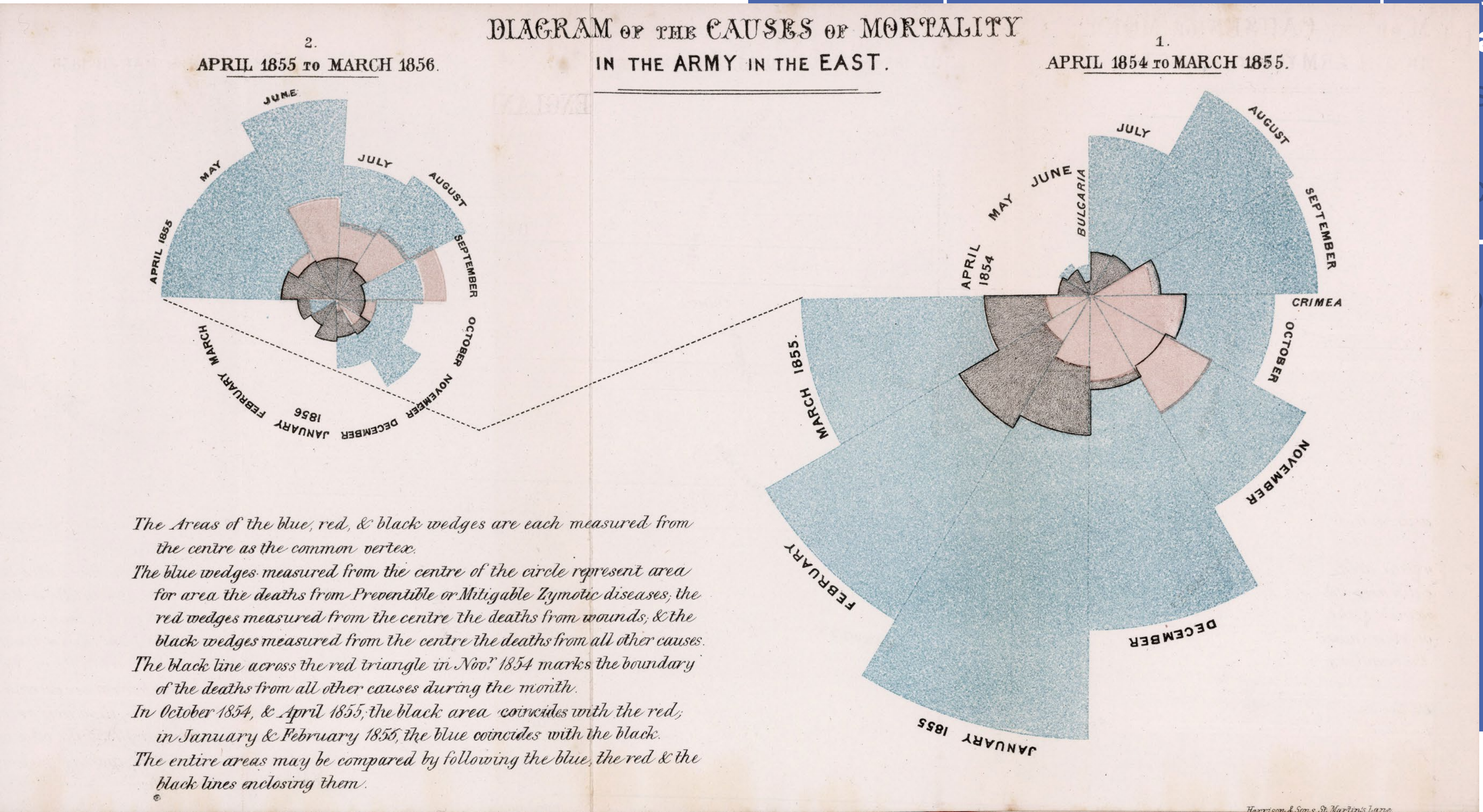
news and government

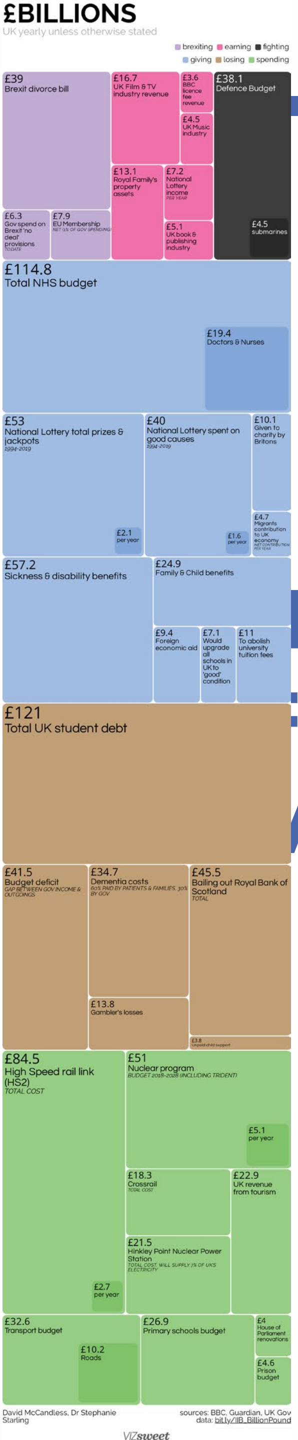
is & Sep Kamvar)

scale

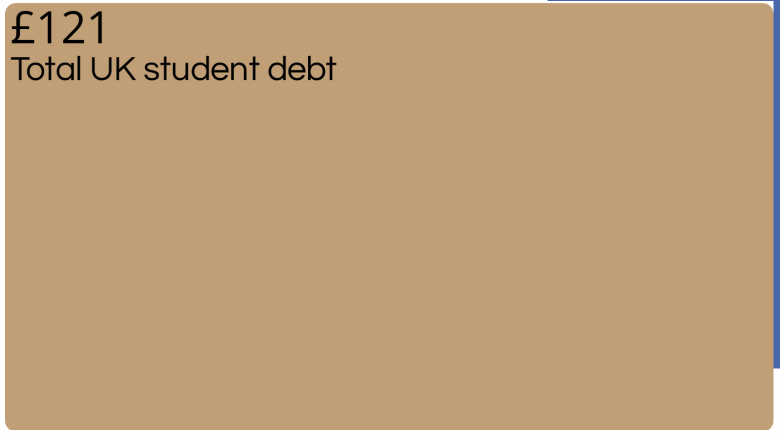


- **Crimespotting (Stamen Design)**
- **Polar Area Graph (Florence Nightingale)**
  - Illustrated deaths from preventable diseases in hospitals
  - Sparked a revolution in hospital hygiene
- **andless)**
  - s in news and government
  - ctive
- **Harris & Sep Kamvar)**
  - bal scale
  - ngs





IE POWER
F VISUAL
ATISTICS



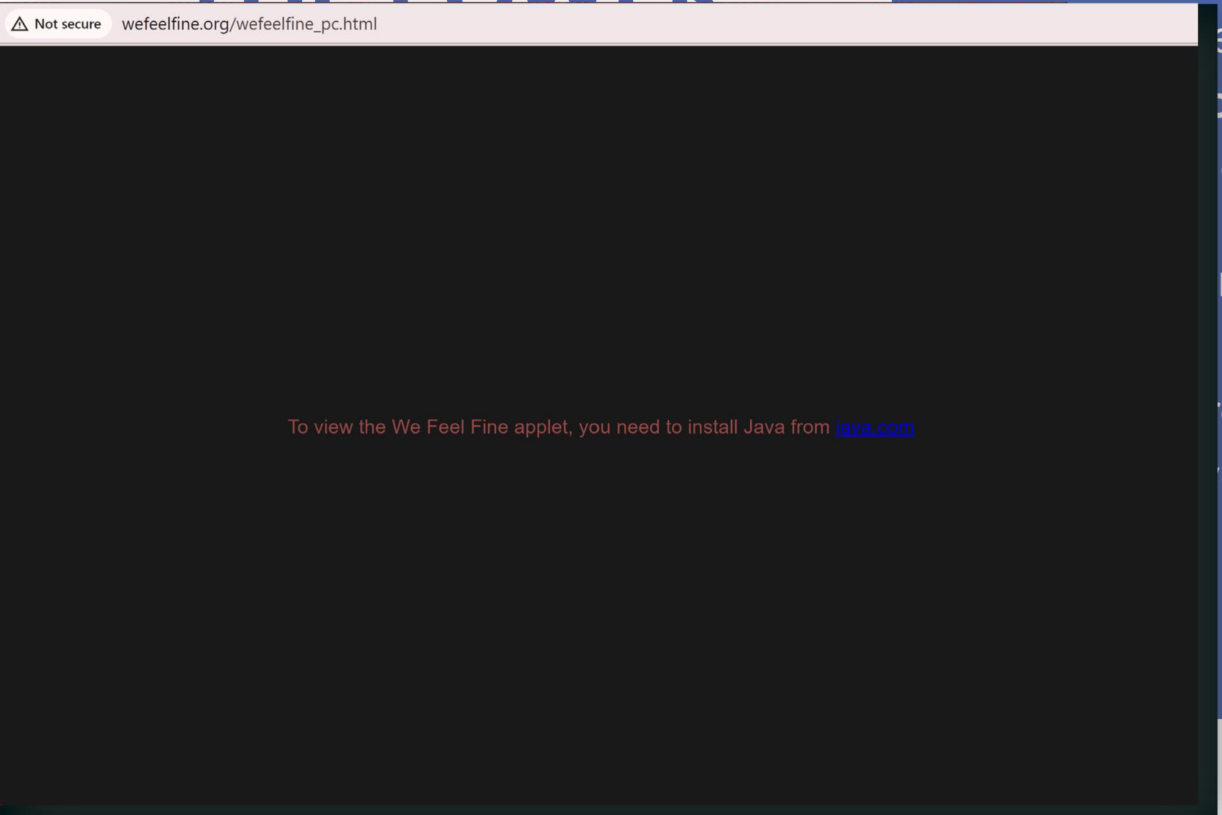
- **Crimespotting (Stamen Design)**
- **Polar Area Graph (Florence Nightingale)**
- **Billion Pound-O-Gram (David McCandless)**
  - Helps comprehend large numbers in news and government reports
  - Puts financial figures into perspective
- **"We Feel Fine" Project (Jonathan Harris & Sep Kamvar)**
  - Explores human emotion on a global scale
  - Uncovers trends in collective feelings

https://informationisbeautiful.net/visualizations/the-billion-pound-o-gram/

https://jjh.org/we-feel-fine

THE POWER

- **Crimespotting (Stamen Design)**
    - Visualizes relationship between topography and crime
    - Empowers citizens with information for accountability
    - Showed the “promise of visualizations of public data”
- **Polar Area Graph (Florence Nightingale)**
    - Illustrated deaths from preventable diseases in hospitals
    - ed a revolution in hospital hygiene
- **ound-O-Gram (David McCandless)**
    - comprehend large numbers in news and government
    - nancial figures into perspective
- **Fine" Project (Jonathan Harris & Sep Kamvar)**
    - es human emotion on a global scale
    - ers trends in collective feelings



THE POWER OF VISUAL STATISTICS

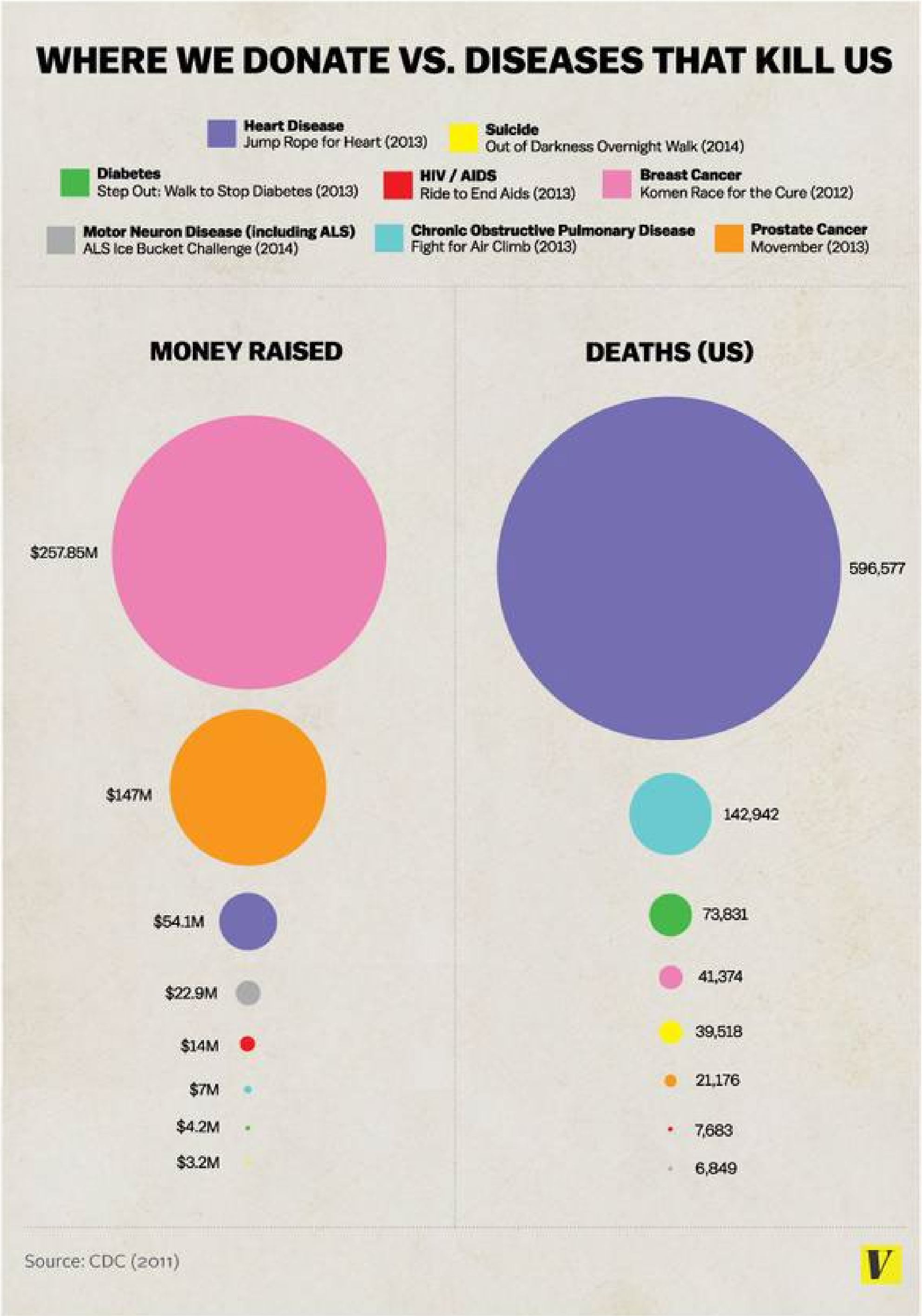
- **Crimespotting (Stamen Design)**
  - Visualizes relationship between topography and crime
  - Empowers citizens with information for accountability
  - Showed the “promise of visualizations of public data”
- **Polar Area Graph (Florence Nightingale)**
  - Illustrated deaths from preventable diseases in hospitals
  - Sparked a revolution in hospital hygiene
- **Billion Pound-O-Gram (David McCandless)**
  - Helps comprehend large numbers in news and government reports
  - Puts financial figures into perspective
- **"We Feel Fine" Project (Jonathan Harris & Sep Kamvar)**
  - Explores human emotion on a global scale
  - Uncovers trends in collective feelings

DATA ARE EVERYWHERE

- We're in a "data-driven" age, right?
- But how do we "consume" these data correctly?
  - How do we get meaning from these data?
- Statistic
  - a number computed from a sample of data

STATISTICS ARE EVERYWHERE

- Education
  - SAT, GPA
- Medicine
  - Blood Cell Count, Cholesterol, Weight
- Public Health
  - Positivity Rates
  - Case Counts
- Sports
  - Baseball, Cricket
- Politics
  - Polling



STATISTICS CAN "LIE"

- The average individual has one testicle and one ovary
- Statistics can be used to mislead people
  - Money and happiness
- "Lie?"

STEVEN PINKER

"..IF NARRATIVES WITHOUT STATISTICS ARE BLIND, STATISTICS WITHOUT NARRATIVES ARE EMPTY."

QUESTION

WHAT DO YOU GAIN FROM BEING A CRITICAL CONSUMER OF RESEARCH?

CRITICAL THINKING STARTS WITH STATISTICS

- What I, in my complacency, chose to ignore is just how much of the persuasion now aimed at the average citizen comes in the form of numbers, specifically numbers that tell us about the future, about how likely something is to happen (or not happen) based on how much it happened (or didn't) in the past. These numbers sing to us the siren song of cause and effect, humanity's favorite tune. Why do we like it so much? **Because knowing what causes events and conditions is the first step toward controlling them, and we human beings are all about controlling our environments**. That's how we ended up ruling this planet, and it's how some of us hope to save it.

Laura Miller, Slate, August 31st 2015

PSYCHOLOGY!

PSYCHOLOGY AS A WAY OF THINKING

WHY IS PSYCHOLOGY SO POPULAR?

Concerned with understanding the way people behave, feel, and think

POPULAR THEMES

- free will and determinism
- nature and nurture
- love and loss
- personality (growth and disorder)
- justice and violations

WHAT DO SCIENTISTS DO?

Scientists are empiricists.

Scientists test theories: the theory-data cycle.

Scientists work in a community.

Scientists tackle applied and basic problems.

Scientists make their work public.

Scientists talk to the world: from journal to journalism.

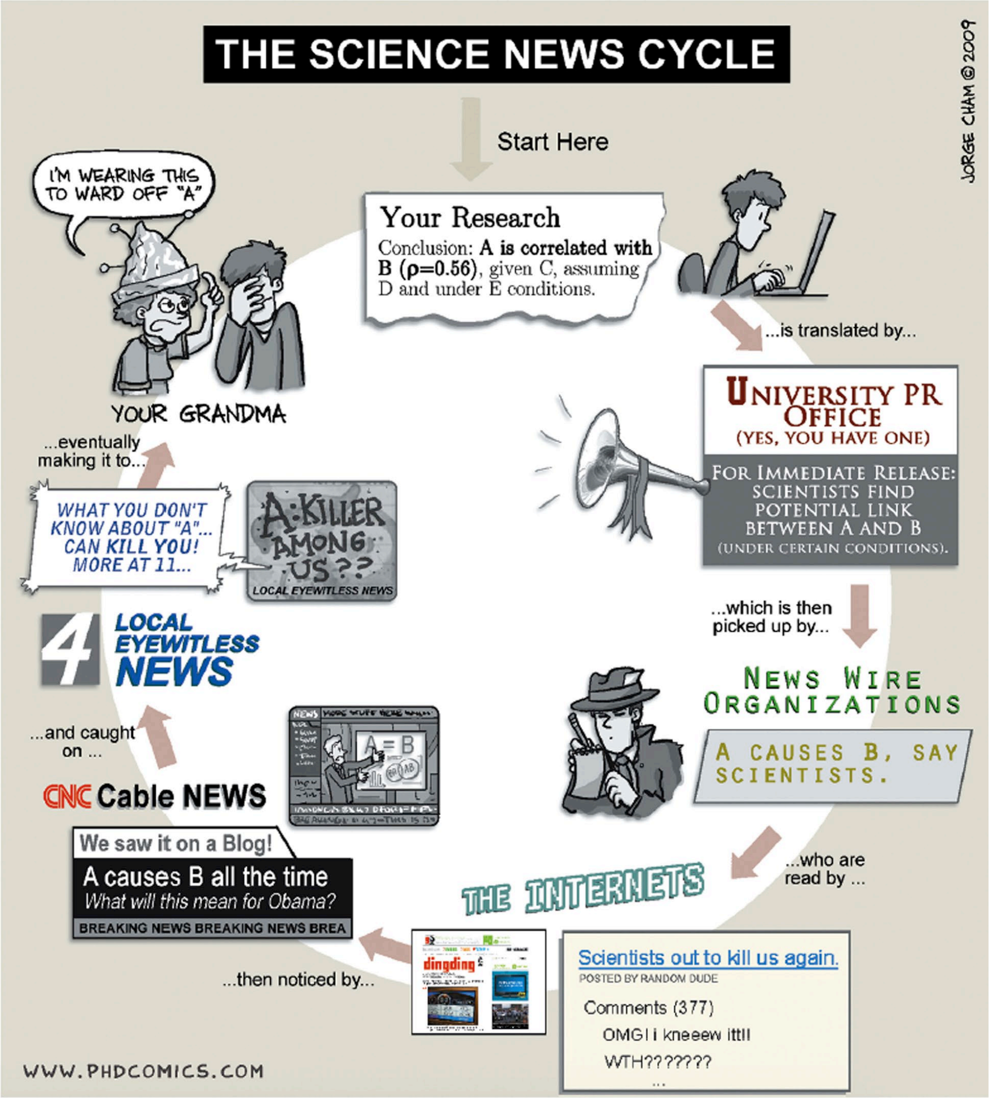


FIGURE 1.9

PHD Comics.com

WRAPPING UP…

THIS TIME…
RESEARCH PROCESS

STEPS IN A RESEARCH STUDY

- Question
- Design
- Sampling
- Measurement
- Summary

SCIENCE STARTS WITH A QUESTION

ASKING A GOOD QUESTION

Research Problem

Research Question

Operationalize

Hypothesize

GOOD QUESTIONS BEGIN WITH PROBLEMS

- Research problems are broad issues or situations that we'd like to
  - describe,
  - understand, or
  - change.

RESEARCH PROBLEMS

- Broadly, these include:
  - Areas of concern,
  - Conditions that could be improved,
  - Difficulties that need to be eliminated, and
  - Questions seeking answers.

‘TOP TEN’ PROBLEMS IN SOCIAL SCIENCE

- How can we induce people to look after their health?
- How do societies create effective and resilient institutions, such as governments?
- How can humanity increase its collective wisdom?
- How do we reduce the ‘skill gap’ between black and white people in America?

Giles (2011) Nature

‘TOP TEN’ PROBLEMS IN SOCIAL SCIENCE

- How can we aggregate information possessed by individuals to make the best decisions?
- How can we understand the human capacity to create and articulate knowledge?
- Why do so many female workers still earn less than male workers?
- How and why does the ‘social’ become ‘biological’?

Giles (2011) Nature

‘TOP TEN’ PROBLEMS IN SOCIAL SCIENCE

- How can we be robust against ‘black swans’ — rare events that have extreme consequences?
- Why do social processes, in particular civil violence, either persist over time or suddenly change?

Giles (2011) Nature

RESEARCH QUESTIONS

A research problem leads to a research question

Questions should in some way . . .

Be worth investigating,

Contribute knowledge to the field,

Improve educational practice, or

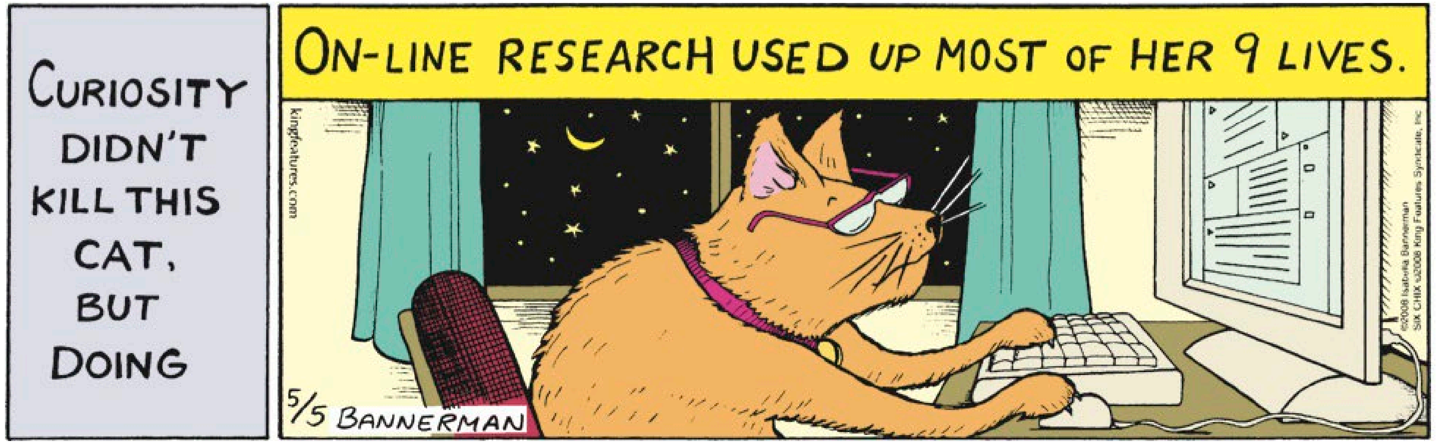
Improve the human condition.

ASKING GOOD QUESTIONS

Begin by	Begin by asking several questions about a topic.
Eliminate	Eliminate questions that cannot be answered by direct observation or by gathering evidence.
Break	Break broad questions into smaller questions that can be investigated one at a time.

RESEARCH QUESTIONS

- Characteristics of a good research question:
  - The question is feasible.
  - The question is clear.
  - The question is meaningful.
  - The question is ethical.



ASKING SCIENTIFIC QUESTIONS

"What is the relationship between . . . ?"

"What factors cause . . ?"

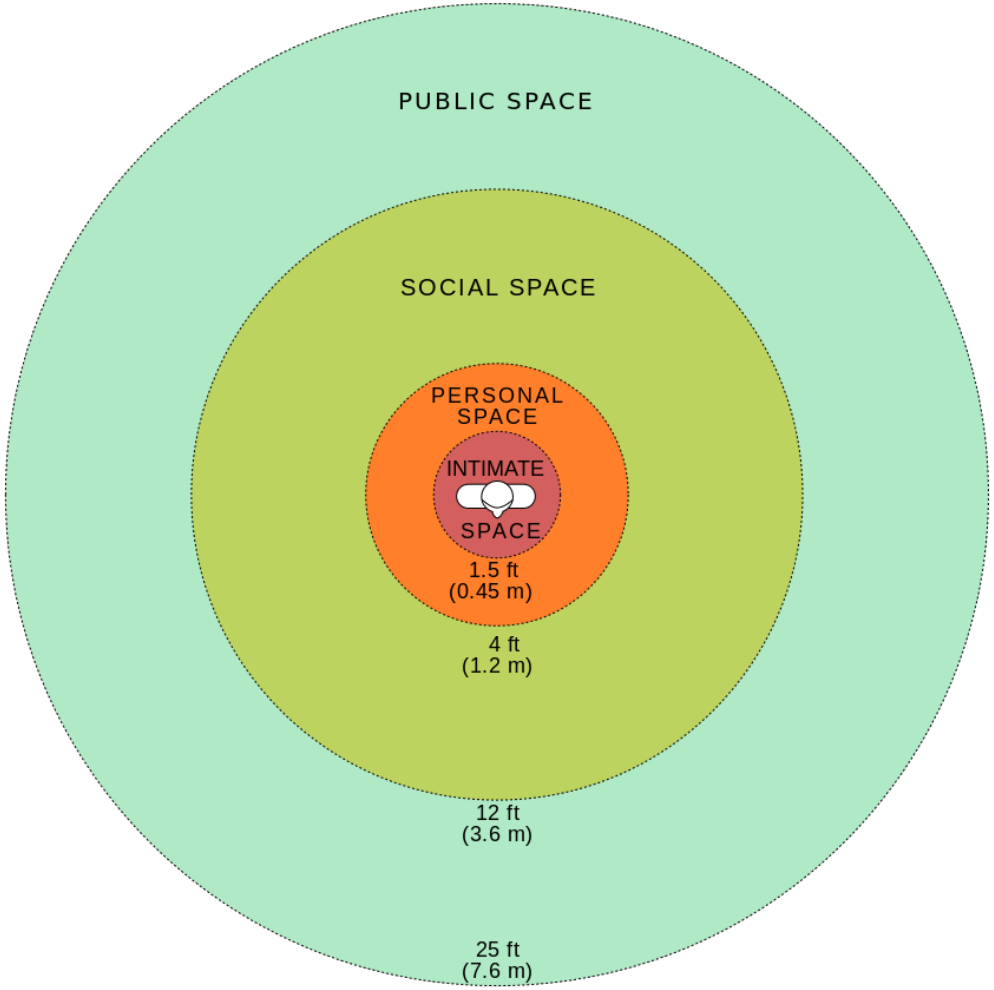
"What is the effect of . . ?"

OPERATIONALIZE

- Operationalization is a process of defining the fuzzy concepts; and
- making the concept(s)
  - distinguishable,
  - measurable, and
  - understandable in terms of observable data.
- Determine which variables you would like to understand.
- You will want to be specific.

EXAMPLE: PERSONAL SPACE

- Personal Space
  - Public Space (25 ft)
  - Social Space (12 ft)
  - Personal Space (4 ft)
  - Intimate Space (1.5 ft)



- We'll delve deeper into measuring later in the class

HYPOTHESIZING

A hypothesis is a

precise, testable statement of

what the researchers predict will be the outcome of the study.

In other words:

What do you think the answer is,

- given what we already know?

Research Problem:

What are the causes of health inequity?

Research Question:

Specifically, what is the effect of social stress on health, among naturally anxious individuals?

EXAMPLE

EXAMPLE

- Operationalize:
  - Stress: Entering Person's Intimate Space (1.5 ft)
  - Health: Self-report ailment checklist
  - Naturally Anxious: Self-reported scores of Neuroticism (Top 20%)
- Hypothesis:
  - Entering a person's intimate space will cause individuals to check more ailments on the self-report checklist.

WRAPPING UP…